

Henrique Teixeira Kalke

Senior Software Engineer · Fintech & Banking Systems · Artificial Intelligence

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PROFESSIONAL SUMMARY

Senior Software Engineer with 6+ years of experience building mission-critical financial systems where robust engineering meets complex regulation. Career spanning accountingtech, credit recovery fintech, and capital markets infrastructure — always at the intersection of high-scale systems, financial domain knowledge, and automation. Currently building the pillars of a complete banking ecosystem at Kanastra: automated regulatory onboarding (KYC / Due Diligence), boleto issuance and settlement, PIX, automated fee billing, and AI-powered document validation using LLMs. Strong background in event-driven microservices (Python, TypeScript, Apache Kafka), cloud infrastructure (GCP, AWS), and Central Bank of Brazil (Bacen) regulatory compliance.

SKILLS

Languages: Python, TypeScript, JavaScript, Go, SQL

Backend: FastAPI, NestJS, Node.js, Django, Flask, REST APIs, Microservices, Event-Driven Architecture

AI & Machine Learning: LLMs, LangChain, OpenAI API, Google Vertex AI, Google Document AI, NLP, TensorFlow, Keras

Data & Messaging: PostgreSQL, MySQL, Redis, Apache Kafka, RabbitMQ, Celery, Prisma ORM

Cloud & DevOps: Google Cloud Platform (GCP), AWS, Docker, Kubernetes, ArgoCD, GitHub Actions, CI/CD, OpenTelemetry, Grafana, SonarQube

Frontend: React, Vue.js, Tailwind CSS, Vite

Financial Domain: PIX, Boleto (CNAB), TED, KYC / Due Diligence, Bacen Regulatory Compliance

PROFESSIONAL EXPERIENCE

Senior Software Engineer — Kanastra (Kanastra Banking) May 2024 – Present · Remote, Brazil

Kanastra is the technology infrastructure for capital markets and banking services for investment funds. I work on the banking vertical, building critical banking systems under high technical and regulatory complexity.

- Conceived, designed, and led the implementation of a fully automated account onboarding flow with Due Diligence and Bacen regulatory compliance — eliminated the entire manual process, grew the customer base by 30%, and ensured every enabled account had immediate access to PIX, Boleto, and TED.
- Build and maintain AI-powered document validation and parsing services using Python (FastAPI), LangChain, OpenAI, and Google Document AI / Vertex AI, with Kafka-based event pipelines that automate the analysis of regulatory documents in customer onboarding.
- Helped build and currently maintain Kanastra's complete boleto infrastructure (CNAB, registration, and settlement), supporting 1,000+ boletos issued per day and over R\$ 1 million in daily settled volume.
- Developed a fully automated per-account fee-billing engine, creating a new recurring revenue stream for the company with zero operational intervention.
- Develop event-driven microservices in TypeScript (NestJS, Prisma, PostgreSQL, Kafka, Redis) and React frontends, deployed on GCP with Docker, Kubernetes (ArgoCD), and GitHub Actions CI/CD.
- Implemented Single Sign-On (SSO) integration in partnership with Itaú, enabling frictionless, institution-grade authentication for users with Kanastra and Itaú institutional accounts.
- Work directly on Bacen regulatory alignment for account opening, transfers, and payment flows, bringing legal and operational security to the entire banking vertical.
- Built from scratch a three-tier (N1 / N2 / N3) banking technical support model, creating processes and playbooks that ensure operational stability for a critical financial product.

Mid-Level Software Engineer — QuiteJá Aug 2023 – May 2024 · Curitiba, Brazil · Hybrid

QuiteJá is a leading fintech in the credit recovery sector, connecting major financial institutions with end consumers.

- Served as the technical reference for the team, conducting rigorous code reviews, delivering structured feedback, and mentoring junior engineers — directly contributing to the team's technical growth.
- Acted as the direct technical interface with creditors' engineering teams, integrating BMG, Agibank, Via Varejo, and Bradesco into the platform — integrations pivotal to the jump from 2M to 6M total paid boletos and the expansion of 3M+ customers.

Junior Software Engineer — QuiteJá

Oct 2021 – Aug 2023 · Curitiba, Brazil · Hybrid

- Developed and maintained integrations with APIs from major creditors such as Santander, Via Varejo, and Porto Seguro, enabling real-time display of debt negotiation offers on the platform.
- Worked extensively with Python, Flask, and Celery with RabbitMQ as the broker, ensuring efficient background task processing and high-volume inter-microservice communication.
- Redesigned the remittance file processing pipeline on AWS ECS, achieving a 15% reduction in cloud costs related to that flow.

Software Engineering Intern — Exati Tecnologia

Feb 2021 – Oct 2021 · Curitiba, Brazil · Hybrid

Exati is a reference in technology for urban infrastructure management and Smart Cities (public lighting telemanagement).

- Developed full-stack features: Vue.js and TypeScript on the front-end for real-time monitoring dashboards, and Python on the back-end for device telemetry processing.
- Implemented and maintained IoT automation flows with Node-RED, integrating hardware devices directly into the management platform — a critical part of the telemanagement architecture.

Machine Learning Engineer — ROIT

Nov 2019 – Sep 2020 · Curitiba, Brazil · Hybrid

ROIT is a pioneering accountingtech in hyperautomation for accounting; I worked on the 'Revolution' innovation team building end-to-end AI solutions.

- Developed facial recognition and image classification features for the ROIT People app using Deep Learning with Keras and TensorFlow — shipped to production and used internally.
- Contributed to DUKA, an intelligent chatbot with Named Entity Recognition (NER) for service automation and data triage, used in production.
- Deployed ML models on Google Cloud Platform (Cloud Run, Cloud Functions) with CI/CD pipelines via Jenkins.

IT Intern — Celepar

Jun 2019 – Nov 2019 · Curitiba, Brazil · On-site

- Maintained the mission-critical infrastructure of HEMEPAR, the state of Paraná's blood bank, ensuring uninterrupted system operation and performing hardware and software troubleshooting on Windows and Windows Server.

Technical Support Intern — ConnectPlug

Jul 2018 – Nov 2018 · Curitiba, Brazil · On-site

- Activated and configured the SAT fiscal module with SEFAZ-SP and provided technical support to retail clients on point-of-sale (POS) systems and peripherals.

EDUCATION

Technologist Degree in Applied Artificial Intelligence — PUC-PR

2019 – 2021 · Curitiba, Brazil

LANGUAGES

- **Portuguese** — Native
- **English** — Advanced · Schooled in London with 4 years of immersion; active daily technical use at work.
- **German** — Basic · 3 modules completed at CELIN.